

A literature comparison matrix

Project: Reproducible AI Research Study

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Replication and reproduction

Without opening the old lesson, explain one key idea from Replication and reproduction. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Finding primary research

Without opening the old lesson, explain one key idea from Finding primary research. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for A literature comparison matrix. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for A literature comparison matrix. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

A has recall@5 .8; B has precision@3 .9 on another corpus. Can you rank them?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

6. W2 Trace the worked case

Study A reports accuracy90% on easy classification; B reports claim support80% on difficult generated answers. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

7. W3 Audit the limit

A report omits this qualification: A matrix documents comparison limits; it does not create missing common evidence. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

8. W4 Apply a new case independently

A has recall@5 .8; B has precision@3 .9 on another corpus. Can you rank them?

My reasoning and evidence

9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled a literature comparison matrix evidence sheet to Reproducible AI Research Study. Show how you can explain the mechanism and terminology of a literature comparison matrix. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week’s record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: A literature comparison matrix

A literature matrix that preserves differences. Two fictional studies evaluate related tools under different conditions.

Study S1: 8/10 correct on indoor photos. S2: 16/20 on outdoor photos. Both report 80%.

AI ACADEMY / AI-7/6

Same rate, different study conditions

	Correct	Total	Condition
Study S1	8	10	Indoor
Study S2	16	20	Outdoor

Read a cell using both its row and column labels.

A literature comparison matrix

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Record each method

Keep population, task, model version and scoring in separate columns.



2. Compare rates carefully

Both rates are 80%, with different denominators and conditions.



3. Name missing comparison

Neither study directly compares both tools on the same photos.

CHANGE ONE THING

Remove the indoor/outdoor column.

Explain the changed result using the matching lesson.

Explain why 8/10 indoors and 16/20 outdoors share a rate but not a controlled comparison. Which condition column must the literature matrix preserve?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

13. Project portfolio milestone

Project: Reproducible AI Research Study

Current milestone: Frame the need. Name the user, the problem, allowed inputs and a safe simple starting method.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A literature matrix that preserves differences. Two fictional studies evaluate related tools under different conditions.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Neither study directly compares both tools on the same photos.

1. Name missing comparison
2. Record each method
3. Compare rates carefully

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Name missing comparison Neither study directly compares both tools on the same photos. A literature matrix helps compare evidence without flattening differences. Equal percentages do not mean equivalent tasks, samples or methods.

Practice 2: explain the deciding evidence

Changed-case practice: Remove the indoor/outdoor column. Which conclusion is supported by the supplied facts?

1. The matrix conceals an important condition difference.
2. We can decide the result without examining the changed input or the procedure.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The matrix conceals an important condition difference. A reader might now mistake unlike studies for a controlled head-to-head comparison.

Repair a missing building block

Review Replication and reproduction (ai-7/5). Goal: Explain the mechanism and terminology of replication and reproduction. Short practice: Calculate the scores from the original and new four-record sets. Which activity reproduces saved analysis, which repeats the protocol, and why does a different rerun of the original data need investigation?. Return to this week and explain what you changed.

Review Finding primary research (ai-7/3). Goal: Explain the mechanism and terminology of finding primary research. Short practice: Separate what the blog claims from what the original report records. Which methods and denominator must be inspected, and what status is honest if the original report is unavailable?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
